Beyond costs: Mapping Norwegian youth perspectives for inclusive energy transition

Modelling Questionnaire

Supplemental file

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Question for all:

According to you, what is the most important to you that should be considered when installing renewable energy technologies? Rank them 1 as the most important and 3 as the least important (i.e., cost, nature reserves).

1

2

3

Table attributes explanation

Household electricity savings: Spending less on home energy because wind and solar power are cheaper now. This can also lead to more local municipality benefits, i.e., sports facilities of people's choice.

Visual impact: How often the installed wind/solar farms will be exposed/seen by local people. There are three categories of visual impact: High (seen by many people); Medium (seen only by a few people); Low (barely seen by anyone).

Height: Height of Wind Turbines/Solar Panels. This can influence how they look in the landscape.

Area needed to provide electricity to a house: It is the area (in square meters) needed to install wind/solar technology to meet the electricity consumption of one Norwegian house.

Power lines requirement: Transmission lines are needed to connect renewable energy farms with the electricity grid. Here three categories are introduced: High (long-distance transmission lines); Medium (short-distance transmission lines); Low (near-to-zero transmission expansion needed).

Q1: Ask ALL

	Off-shore wind 	On-shore wind 
Household electricity savings	0 NOK per month	850 NOK per month
Visual impact	Low	Medium to high
Height	Above 200 meters	Below 200 meters
Area needed to provide electricity to a house (m2)	60	200 to 600
Power lines requirement	High	Medium to high

Q2: If off-shore=1

	Off-shore wind 	 +  Solar + Off-shore wind
Household electricity savings	0 NOK per month	328 NOK per month
Visual impact	Low	Medium to Low
Height	Above 200 meters	Low (solar) + high (off-shore wind)
Area needed to provide electricity to a house (m^2)	60	120 to150
Power lines requirement	High	Medium to high

Q3: if on-shore=1

	On-shore wind large 	On-shore wind small 
Household electricity savings	850 NOK per month	1050 NOK per month
Visual impact	Medium	high
Height	150-200 meters	100-150 meters
Area needed to provide electricity to a house (m2)	270	670
Power lines requirement	Medium to high	Low to Medium

Q4: if on-shore large=1

	On-shore wind large 	 +  Solar + On-shore wind large
Household electricity savings	847 NOK per month	766 NOK per month
Visual impact	Medium	Medium
Height	150-200 meters	Low (solar) + high (on-shore wind)
Area needed to provide electricity to a house (m2)	270	310
Power lines requirement	Medium to high	Medium to high

Q5: if on-shore small=1

	On-shore wind small 	 +  Solar + On-shore wind small
Household electricity savings	847 NOK per month	766 NOK per month
Visual impact	high	Medium to high
Height	100-150 meters	Low (PV) + High (on-shore wind)
Area needed to provide electricity to a house (m2)	670	410
Power lines requirement	Low to Medium	Low to Medium

Questions for all:

Power lines help connect new electric power stations to the local electricity grid. As Norway's electricity needs grow, we'll add more clean-energy power stations. There are two main ways to build these power lines:

Above ground power lines

Below ground power lines
(almost 20 times more expensive)

How would you like new power lines to be built?

- No new power lines should be built
- Power lines above the ground
- Power lines below the ground (cost 20 times more than above-ground lines)
- Power lines below the ground in remote areas, above the ground in cities
- Power lines above the ground in remote areas, below the ground in cities

The weather is the main driver for how much renewable energy (solar/wind) we can produce, meaning we might generate too much energy when we do not need it or not enough when we do need it. Which option do you think is best to ensure we have the right amount of energy at all times?

- Share extra electricity with nearby countries and get extra electricity from these countries when we need it
- Keep extra electricity from renewable energy saved locally, for example, in batteries, to use later
- Use a mix approach and save of electricity locally, while also share it with nearby countries
- Try to use electricity when we make a lot of it (changing habitual patterns)

Currently, Norway is a net electricity exporter; what do you think about the country's future energy supply?

- Export more than it does today
- Export same as today
- Produce enough for its use (net-zero)
- Only import from other countries

Below are some features of an energy system. Please rank them based on how important they are to you. 1 = most important

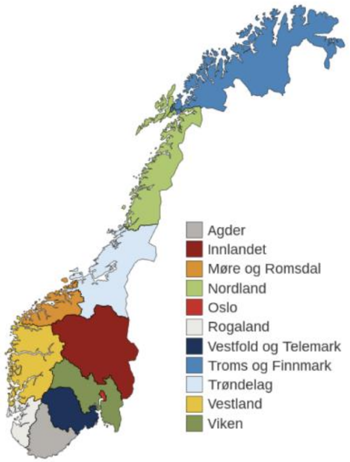
- Zero carbon emissions
- Being able to produce all electricity locally from renewable energy (self-sufficiency)
- Keeping the cost of electricity low
- Protecting plants, animals, and birds (biodiversity)

To what extent do you agree with the following statement?

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
Renewable energy (wind/solar) should be prioritized to meet future energy demand, even if it leads to higher electricity bills for Norwegian people.	☐	☐	☐	☐	☐
Future demand can be met by buying electricity from other countries instead of installing more power stations locally.	☐	☐	☐	☐	☐
Nature reserves (animal/bird protection) must be taken care of in deciding the wind turbine installations.	☐	☐	☐	☐	☐

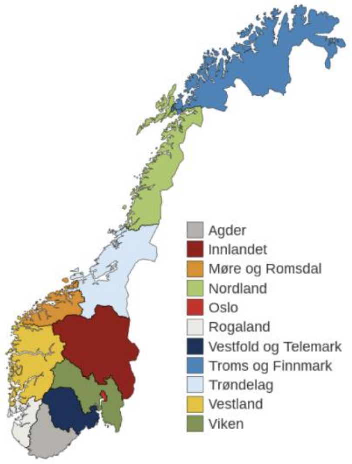
Which Norwegian regions do you think are most appropriate to install wind farms (you may select more than one region):

- ○ Agder
- ○ Innlandet
- ○ Møre og Romsdal
- ○ Nordland
- ○ Oslo
- ○ Rogaland
- ○ Vestfold og Telemark
- ○ Troms og Finmark
- ○ Trøndelag
- ○ Vestland
- ○ Viken



Which Norwegian regions do you think are most appropriate to install solar farms (you may select more than one region):

- Agder
- Innlandet
- Møre og Romsdal
- Nordland
- Oslo
- Rogaland
- Vestfold og Telemark
- Troms og Finmark
- Trøndelag
- Vestland
- Viken



To what extent do you agree to place wind installations in the following locations:

	Strongly disagree	**Disagree**	**Neither agree nor disagree**	**Agree**	**Strongly agree**
In areas where there is little economic activity to attract new industry.	☐	☐	☐	☐	☐
Near electricity consumption places where most people live, i.e., counties/municipalities?	☐	☐	☐	☐	☐
In remote areas with not much electricity consumption (high transmission cost).	☐	☐	☐	☐	☐

In areas where much of the natural landscapes are changed by human activities (cities/towns).	☐	☐	☐	☐	☐
In areas where much of the natural landscape is unaffected by human activities (forests/remote areas).	☐	☐	☐	☐	☐
Build renewable energy power stations near existing roads and power lines to minimize the need for new ones.	☐	☐	☐	☐	☐

Take a look at the pictures of different types of landscapes with and without wind turbines. To what extent do you agree or disagree to install wind farms in the following locations. Remember, there are no correct or incorrect answers. Your opinion is what counts the most.

Bare rock mountains: Tall and steep landscapes with exposed rock

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

Scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Bare rock mountains	☐	☐	☐	☐	☐	☐	☐	☐	☐

Residential/Urban: Part of a town/city where people live.

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

Scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Residential/urban	☐	☐	☐	☐	☐	☐	☐	☐	☐

Commercial/industrial zones: area often filled with factories/warehouses where products are made/stored

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

Scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Commercial/industrial zones	☐	☐	☐	☐	☐	☐	☐	☐	☐

Natural grassland/animal grazing land: wide open space often covered with grasses and some trees

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Natural grassland	☐	☐	☐	☐	☐	☐	☐	☐	☐

Nearshore areas (Beaches/dunes/coastal): Areas near to the sea

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Nearshore areas	☐	☐	☐	☐	☐	☐	☐	☐	☐

Sparsely vegetated areas: Places with not many plants but patches of grass

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Sparsely vegetated areas	☐	☐	☐	☐	☐	☐	☐	☐	☐

Offshore areas: a part of the ocean and far from the land (wind can be installed within a certain distance from the land, depending on how deep the water is)

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
offshore areas	☐	☐	☐	☐	☐	☐	☐	☐	☐

Arable land (agriculture): areas suitable for farming

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
arable land	☐	☐	☐	☐	☐	☐	☐	☐	☐

Forests (Mixed type): large areas covered with different types of trees but also have patches of land

(without wind turbine installation)

(With wind turbine installation)

1: I strongly disagree, 9: strongly agree

scale	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Forests (Mixed type)	☐	☐	☐	☐	☐	☐	☐	☐	☐